Manas Pal

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Profile

Backend Engineer specializing in high-scale, latency-critical ad-tech systems and ETL pipelines. 1+ years shipping production features for a retail-media platform serving 100s of millions of ad requests, with measurable impact on CTR, revenue, and infrastructure cost

Education

Dr. Vishwanath Karad MIT WORLD PEACE UNIVERSITY 2021 – 2025 | Pune, India
B.Tech Computer Science and Engineering
9.02 CGPA

Professional Experience

Osmos by Onlinesales.ai May 2025 – Present | Pune
Software Engineer

- Architected performance-focused backend services and ETL pipelines powering advertising analytics on a platform serving 100s of millions of ad requests across multiple marketplaces, sustaining P95 latency of 100–150ms at peak traffic.
- Designed and shipped a new keyword-to-category prediction pipeline with a fallback path and redesigned Redis storage layout — +0.5 CTR on the pilot client with no RR loss, and ~20% Redis memory reduction across all clients on rollout.
- Led the Performance team's portion of a cross-team Redshift → BigQuery migration for critical pipelines: rewrote queries for BigQuery native execution, refactored ETL workflows, and shipped an automated parity-validation framework diffing on primary keys/emails; in the process, identified and fixed/optimized legacy code issues, driving up to 20% lower pipeline execution time and cloud cost.
- Built a budget-pacing algorithm — including for a new campaign-group (CG) hierarchy layer introduced to support a major enterprise client (PayPal) onboarding — that analyzes real-time traffic trends to throttle spend and prevent budget exhaustion during peak hours, working correctly across both single-campaign and grouped-campaign configurations.
- Built, solo end-to-end, automated monitoring frameworks spanning ETL validation, API monitoring, QA regression, and low-level business logic anomaly detection — catching silent failures invisible to standard service/data monitors (e.g. undersized models, malformed or out-of range inputs, logically incorrect algorithm output) — cutting RCA turnaround for leadership escalations by ~50%; rotate through production on-call, triaging pipeline failures and driving RCAs.
- Leveraged AI-assisted development tools (ChatGPT and Claude) to accelerate debugging, SQL optimization, code reviews, documentation, test-case generation, and solution exploration, improving engineering productivity while ensuring all production code was validated through rigorous testing and peer review.
- Mentor interns and junior engineers on backend development, debugging, and system design.
- Stack: Python, Node.js, Java, Ruby, SQL, BigQuery, Redshift, GCP, AWS, Microservices, Solr, Redis, MongoDB, Git, Linux, OLAP, OLTP, ETL

Osmos by Onlinesales.ai June 2024 – April 2025 | Pune, India
Technical Intern

Started on Infrastructure, transitioned to Performance Engineering.

- Optimized the BigQuery-to-BigQuery fetch-and-store workflow with a MERGE-based strategy in a hybrid Redshift–BigQuery ETL pipeline — ~40% lower BigQuery compute costs and ~50% faster ETL execution.
- Modified a legacy brand-extraction pipeline with new extraction logic and a real-time fallback path — 2× brand coverage, +0.2 avg CTR across major clients, while holding P95 latency at 100–150ms in a latency-sensitive production service.
- Designed and shipped Node.js and Python microservices powering internal data-processing workflows and APIs, and built automated job-orchestration and monitoring for ETL pipelines across BigQuery and Redshift — improving pipeline reliability and cutting manual intervention.

Projects

Hate Speech Detection in Hindi language October 2023 – December 2023

- Fine-tuned **DistilBERT** on the HASOC dataset for Hindi social-media sentiment classification, achieving **84% accuracy**.
- Built training pipeline in PyTorch; served predictions through a React front-end.

Aircraft Blade Defect Inspection (Bharat Forge, ₹10L project) January 2023 – January 2024

- Annotated **3,000+ defects across 500 images** and benchmarked detection architectures; selected **Mask R-CNN** (mAP 0.55–0.59, meeting industry benchmarks).
- Built Flask inference APIs and a React inspection UI; integrated camera SDK (zoom, autofocus, capture).

Skills

— **Programming Languages:** Python, Java, JavaScript, C++, PHP, Ruby, SQL, Bash, HTML, CSS | **Backend & Frameworks:** Node.js, Flask, FastAPI, REST APIs, Microservices | **Databases & Storage:** MySQL, MongoDB, Redis | **Search & Data Platforms:** Apache Solr (Solr), Google BigQuery, Amazon Redshift | **Data Engineering:** ETL Pipelines, Data Warehousing, OLAP, OLTP | **Cloud & DevOps:** Amazon Web Services (AWS), Google Cloud Platform (GCP), Docker, Kubernetes, CI/CD, Git, Linux, Vim | **AI/ML & Data Science:** Machine Learning, Deep Learning, Natural Language Processing (NLP) | **AI-Assisted Development Tools:** ChatGPT, Claude, GitHub Copilot, Google Gemini

Certificates

Machine Learning Specialization
Issued by: DeepLearning.AI, Stanford University

Responsive Web Design
Issued by: freeCodeCamp

Cognizant - Artificial Intelligence Job Simulation
Issued by: Forage